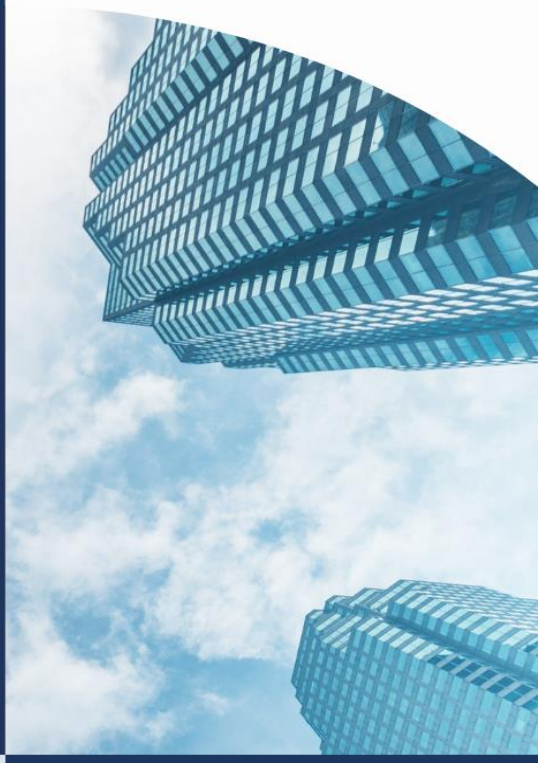
BGIS DEAKIN UNIVERSITY

WORKFORCE OPTIMISATION PROJECT

MIS399: APPLIED BUSINESS PROJECT



PREPARED BY: BW1



OCT 2026

Executive Summary

BGIS engaged Group BW1 to investigate how recruitment, retention, onboarding, scheduling and utilisation are affecting operating margin across its facilities management workforce. Working across three analysis streams including Vacancies & Pay, Workforce Underutilisation, and Capability Gaps, the team cleaned and linked five raw operational datasets covering roughly 2,500 employees and over 430,000 activity records, then used the resulting master file to surface three headline findings:

- A 3.7x surge in terminations in Q1 2026 against the 2025 quarterly average
- $11.36 million in non-billable cost against 26.3% of activity lines carrying no recorded billing status
- A concentration of workload, overtime and non-billable cost in the ELEC and PLUMB trades

The most material discovery is that the non-billable cost problem is not diffuse: 69.6% of the entire non-billable total traces to a single client account (Australian Defence Force, NNSW), and 99.83% of non-billable cost company-wide comes from Reactive Maintenance (RM) work rather than Planned Preventative Maintenance (PPM). This reframes the problem from "BGIS has a broad billing discipline issue" to "BGIS has a specific contract-and-conversion issue that, once fixed, resolves most of the company-wide gap as a side effect."

The report recommends a three-stage, sequenced response: first, review and enforce the Defence-NNSW and BAE Systems contract and billing practices (Weeks 1-4); second, standardise naming conventions, connect fragmented identifiers and automate data-quality checks across the five source systems (Months 1-2); and third, deploy a live HR Interactive Dashboard on BGIS's existing IFS Cloud and Power BI environment for ongoing visibility (Months 2-6). Progress will be tracked through KPIs including reducing the non-billable share from 42.6%, reducing missing billing status from 26.3%, and increasing the financial match rate from 72.5%.

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1. Introduction & Project Background

1.1 Client & Context

BGIS is a global integrated facilities management (IFM) provider operating in a thin-margin, labour-intensive industry, where workforce costs are the single largest lever on profitability.

Australia is one of the most mature IFM outsourcing markets in the Asia Pacific region, with competition increasingly shaped by service quality, technology and data capability rather than price alone. The Australian IFM industry is valued at approximately $12.0 billion in 2025-26, across around 125 operating businesses (Cooke, 2026). The market has a concentrated top tier of large operators alongside a fragmented group of smaller providers, while clients increasingly favour integrated service delivery. This competitive environment places greater emphasis on efficient service delivery and workforce capability.

Workforce shortages remain an industry-wide challenge, particularly across electrical, mechanical and building-services trades, which are relevant to BGIS's operations (Jobs and Skills Australia, 2025). At the same time, industry research identifies a gap between the collection and effective use of operational data, with fragmented systems and underutilised data limiting workforce visibility and decision-making (IFMA IT Community et al., 2024).

The client briefly asked the team to examine how workforce dynamics, such as recruitment and vacancies, retention, onboarding, scheduling and utilisation, are affecting margin, and to translate that analysis into actionable and evidence-based recommendations.

1.2 Project Objectives & Scope

The project was structured around three parallel analysis streams, each assigned to a subgroup with a rotating coordinator role:

- **Vacancies & Pay**: quantifying overtime cost and termination patterns to identify where retention or resourcing support is needed.
- **Workforce Underutilisation**: identifying the main sources of non-billable cost and the scale of recovery opportunity.
- **Capability Gaps**: developing targeted workforce plans for high-pressure trades using overtime and tenure patterns.

Deliverables followed a formal client-engagement structure: a Team Project Plan, Presentation 1 (data quality and initial findings, 27 August 2026), Presentation 2 (further findings and recommendations, 17 September 2026), and this Final Project Report.

1.3 Team Structure

Group BW1 comprised seven student consultants: Lynn Le, Shenal Nathan Witharana, Thuong Luong, Divya Hasandi Hewawickrama, Jade Dang, Yasiru Dinara Abeyawardena and Quynh Hoang, organised into subgroups mapped to the three analysis streams above.

2. Methodology

2.1 Data Sources

Five raw operational datasets were supplied by BGIS and cleaned without altering the original source files:

Table 1. *Data Sources and Analytical Roles*

Dataset	Grain	Role in analysis
Astea Utilisation Data	Activity row (~ 434,447 rows)	Central activity table; hub of the master merge
Employee Pay Report	Employee (Active + Inactive sheets)	Person-level pay basis and roster reference
Financial Demands	Demand-person line	Cost and price fields per activity
Technician Type Costing	Reference table	Standard cost lookup by SA Type ID
Price Book Data	Reference table (~53,000 rows)	Candidate pricing lookup by customer/activity

2.2 Data Quality Methodology

A structured, three-stage data quality methodology (Detect, Standardise, Validate) aligned to ISO/IEC 25012 data quality standards was applied consistently across all five sources. Raw values were always preserved alongside standardised fields so that any cleaned value could be traced back to its original source. Identifiers were cast to text to prevent leading zeros and inconsistent Excel number formats from silently breaking joins, and dates were parsed with explicit Excel serial-date handling rather than default parsing.

The team's governing principle throughout was to flag, not fix: data anomalies were logged in a Data Quality & Assumptions Log and flagged for client confirmation rather than silently corrected. For example, "Unknown" Trade values were retained as a genuine data quality category rather than inferred from technician type or job title, and cost/price fields from Financial Demands were kept as reported rather than labelled "margin" without Finance validation.

2.3 Join Architecture & Validation

The five datasets link through the following key structure, with Astea Utilisation as the central hub:

- Employee Pay ↔ Astea Utilisation (via Person Number / Employee_ID, standardised to a text person_id).
- Astea Utilisation ↔ Financial Demands (via Demand ID + Person Number, the safest link because a single demand can involve multiple people).
- Astea Utilisation ↔ Technician Type Costing (via SA Type ID).
- Financial Demands / Activity ↔ Price Book Data (via Customer ID + Demand Type + Activity Code, a candidate key requiring effective-date selection to resolve to a specific price).

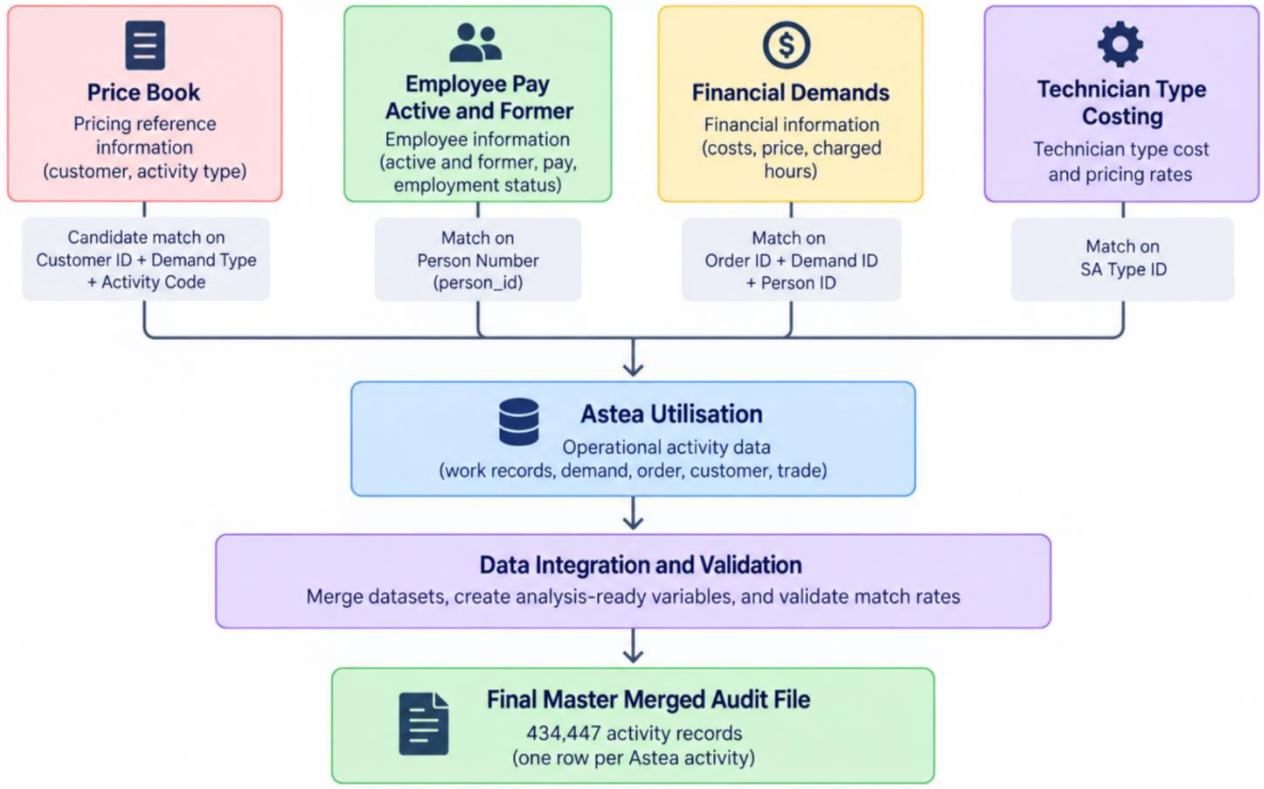


Figure 1. *Data Integration and Validation Process*

Match rates were reviewed before any joined data was interpreted, since a high match rate confirms identifier alignment but not business-meaning equivalence:

Table 2. *Dataset Relationship Match Rates*

Relationship	Rows	Match rate
Utilisation → Financial (Order + Demand + Person)	434,447	72.5%
Utilisation → Technician Costing (SA Type)	434,447	100.0%
Activity/Financial → Price Book (candidate key)	434,447	61.2%

Three audit outputs were produced to keep the analysis transparent and reproducible (referred to Appendix): MIS399_master_merged_audit.csv (the activity-row-level master file), relationship_quality.csv (match percentages across all relationships), and unmatched_utilisation_to_financial.csv (unique keys with no financial match, retained for investigation rather than discarded). All findings in Section 4 are drawn from this master file.

3. Findings

3.1 Vacancies & Pay: Termination Surge

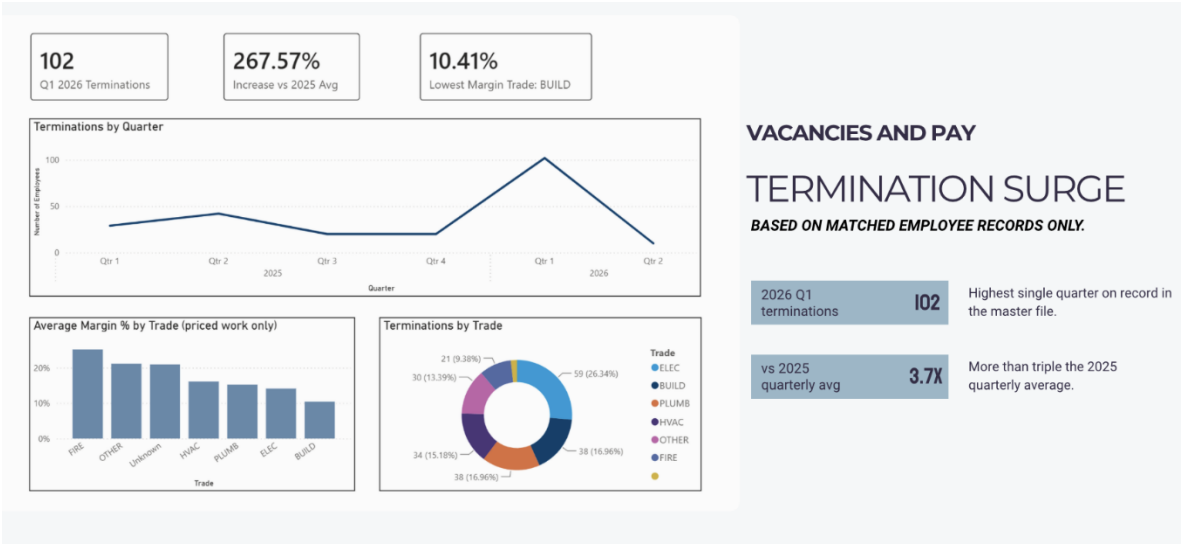


Figure 2. *Vacancies and Pay findings*

Terminations rose sharply from a 2025 quarterly average of roughly 70-79 to 102 in Q1 2026 (more than triple the prior year's typical rate). This directly contradicts internal messaging describing the period as having the "lowest turnover in a decade" and, if corroborated with unmatched records outside the current linkage, represents a high-value finding for BGIS's retention strategy. Because the figure is based on matched employee records only, the true company-wide figure may differ once the 350 of 749 people currently unmatched to the Employee Pay Report are resolved (Section 4.4).

3.2 Workforce Utilisation: Non-Billable Cost

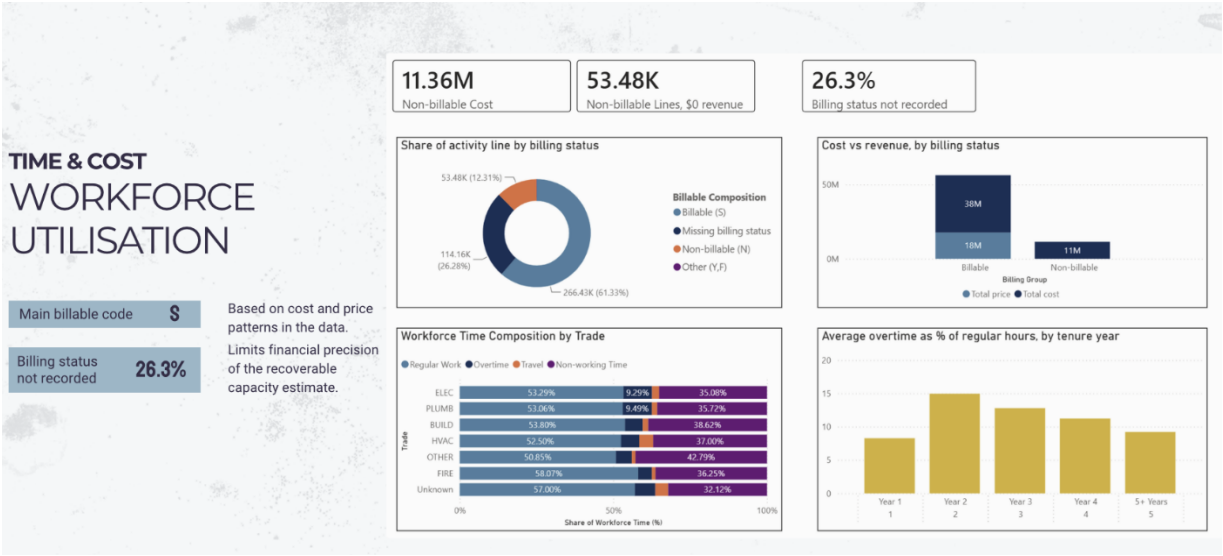


Figure 3*. Workforce Utilisation findings*

Non-billable cost of $11.36 million represents a substantial share of total workforce cost, and over a quarter of all activity lines carry no recorded billing status at all, a gap that limits the financial precision of any recoverable-capacity estimate built on this data. The main billable code identified from cost and price patterns is "S"; billing status is otherwise inconsistently captured at the point of job closure.

This finding was a particular valuable area for further investigation, with the client specifically encouraging the team to develop the unbillable overtime insight beyond the initial company-wide figures. In response, the analysis was extended beyond overall non-billable cost to examine its concentration by trade, client account and work type. This refinement led to the more targeted findings presented in Sections 3.3 and 3.4.

3.3 Capability Gaps: Trade & Grade Analysis

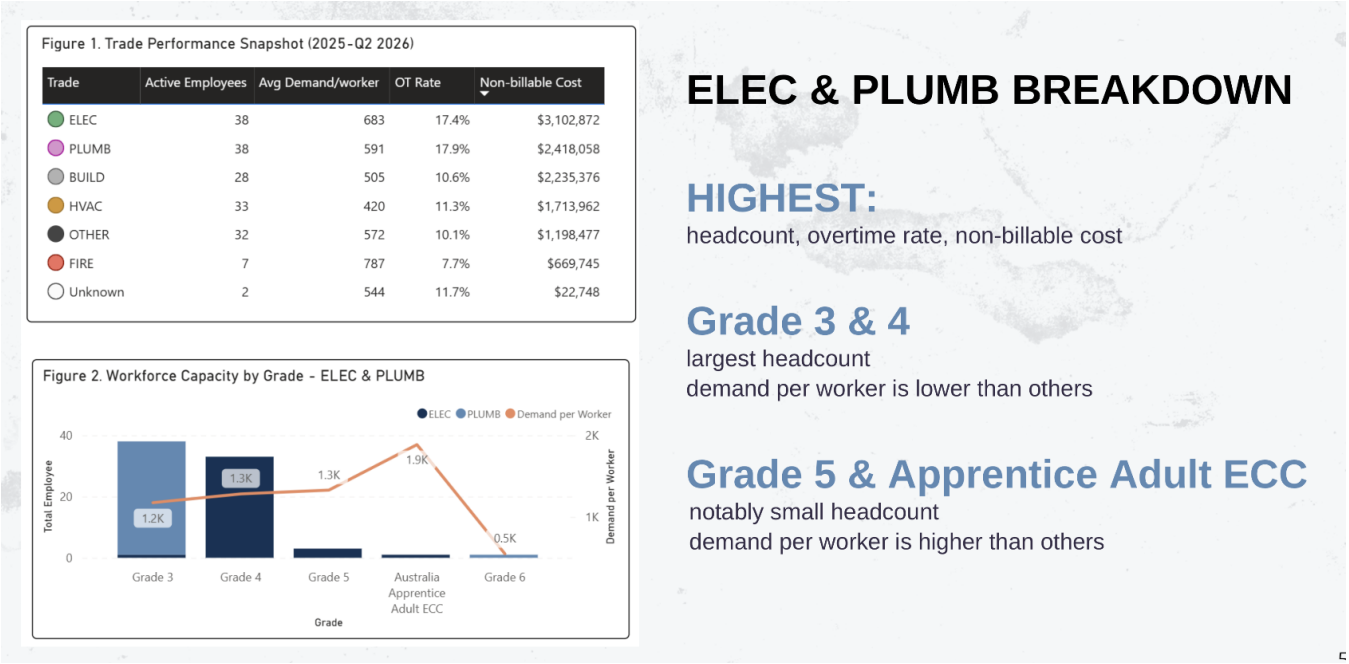


Figure 4. *ELEC & PLUMB analysis*

ELEC and PLUMB lead the company on nearly every workforce metric examined: headcount, overtime rate and non-billable cost are all highest in these two trades. Within them, Grade 3 and Grade 4 technicians represent the largest headcount but a lower demand-per-worker than other grades, while Grade 5 and Apprentice Adult ECC technicians are a notably small headcount group carrying a disproportionately higher demand-per-worker. This combination of high aggregate pressure concentrated in a small number of trades, with an uneven grade-level load which is the basis for the targeted workforce planning recommended in Section 5.

3.4 Root-Cause Findings: Client Account Concentration

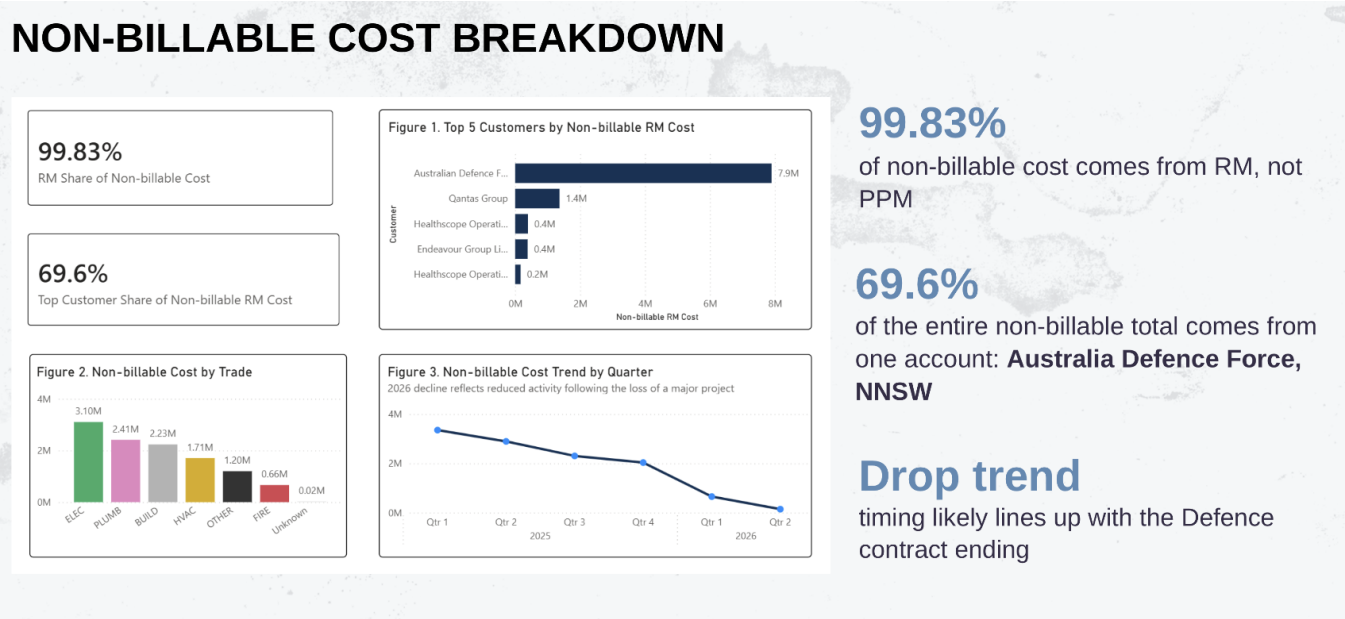


Figure 5. *Non-billable cost breakdown*

The most actionable finding of the project is that the non-billable cost problem is highly concentrated rather than systemic (Figure 4).

A declining trend in this non-billable cost likely lines up with the timing of the Defence contract ending, which would need to be confirmed directly with BGIS before being treated as a resolved issue. Separately, data-quality investigation found that 114,157 rows previously counted as "unmatched" against Financial Demands are in fact Private Activities which is a distinct category that was being miscounted as a genuine matching failure rather than tagged and excluded at entry. Correcting this materially changes the true scale of the matching problem and is addressed directly in the Stage 2 recommendation.

4. Recommendations

Recommendations are sequenced deliberately: fix what is actively bleeding money now, make the underlying data trustworthy next, then build the ongoing visibility needed to catch problems like this earlier in future. Each stage is staged according to the progression of baseline data quality, so later stages are not attempted until the data they depend on is reliable enough to support them.

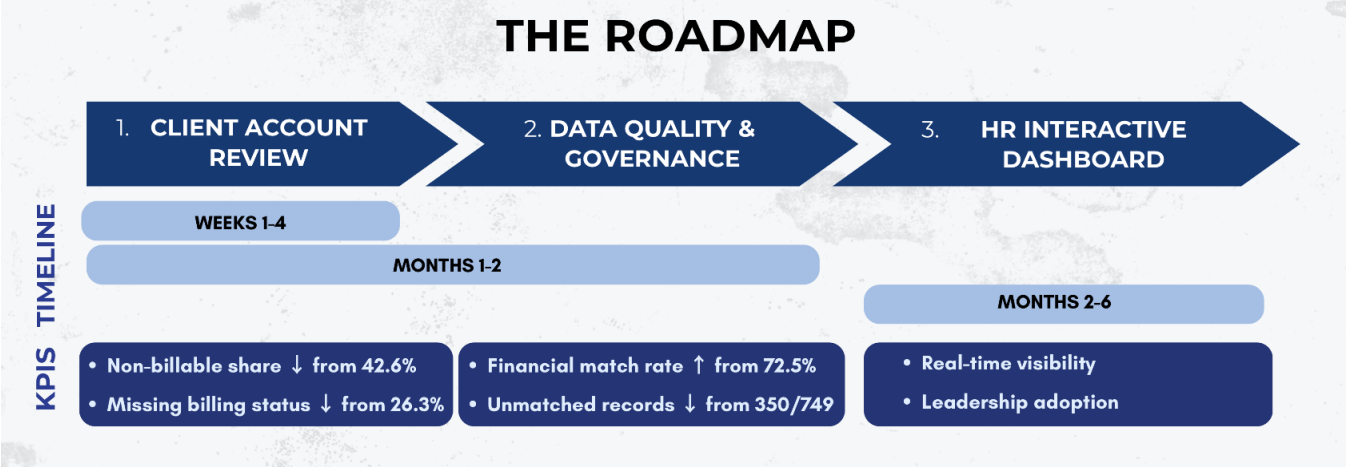


Figure 6. *The recommendation roadmap*

Staging recommendations to the progression of baseline data quality, rather than presenting all three as equally ready to execute, was flagged in Presentation 2 feedback as a mature, practical approach and is carried through unchanged into this report.

4.1 Stage 1: Client Account Review (Weeks 1-4)

Table 3. *Client Account Review Actions*

Action	Detail
Review	Audit the root cause of the Defence-NNSW non-billable concentration before assuming a billing-process fix alone will solve it.
Negotiate	Introduce callout/emergency clauses into the BAE Systems contract (Healthscope, BAE) to formalise how reactive work outside standard scope is billed.
Enforce	Enforce contract terms with Defence-NNSW alongside Finance; apply billing-status tagging at job closure and consistent RM-to-PPM conversion criteria. This same fix also closes the 26.3% company-wide missing-billing-status gap, since the largest contributor to that gap sits within this account.

4.2 Stage 2: Workforce Data Quality & Governance (Months 1-2)

Table 4. *Data Quality and Governance Actions*

Action	Detail
Standardise	Adopt one naming convention across all teams; standardise SA Type, department and billing taxonomies; tag Private Activities at entry so they are never miscounted as unmatched again.
Connect	Migrate legacy employee IDs to the payroll ID scheme; investigate the 5,230-row sync gap; expose Price Book Header/Line IDs so a specific price - not just a candidate match - can be resolved.
Automate	Confirm the Q1-Q2 2026 cost decline with BGIS directly rather than assuming its cause; build ongoing missing-field and ID validation checks into the data pipeline.

4.3 Stage 3: HR Interactive Dashboard (Months 2-6)

The final stage delivers a continuous, live view of workforce performance built on BGIS's existing systems rather than a new platform: BGIS's IFS Cloud environment (the field service source system, using pre-built FACT_/DIM_ data models, estimated at approximately 5 hours of configuration) feeds Power BI (four live views) via an Analytics API, giving leadership real-time, self-service reporting. This requires BGIS's existing IFS Cloud environment, a Power BI workspace on a Pro or Premium licence, Azure AD for secure authentication, and a data gateway. All standard components BGIS is expected to already hold or be able to provision without new platform investment.

The dashboard is organised into four pages, each built to support a distinct type of management decision rather than simply reproducing the same metrics from different angles.

Breaking the dashboard into four areas mapped directly to BGIS's original business problems, and being explicit about what it cannot yet answer, were both singled out in Presentation 2 feedback as strengths - this report preserves both rather than presenting the dashboard as a finished, all-answering solution.

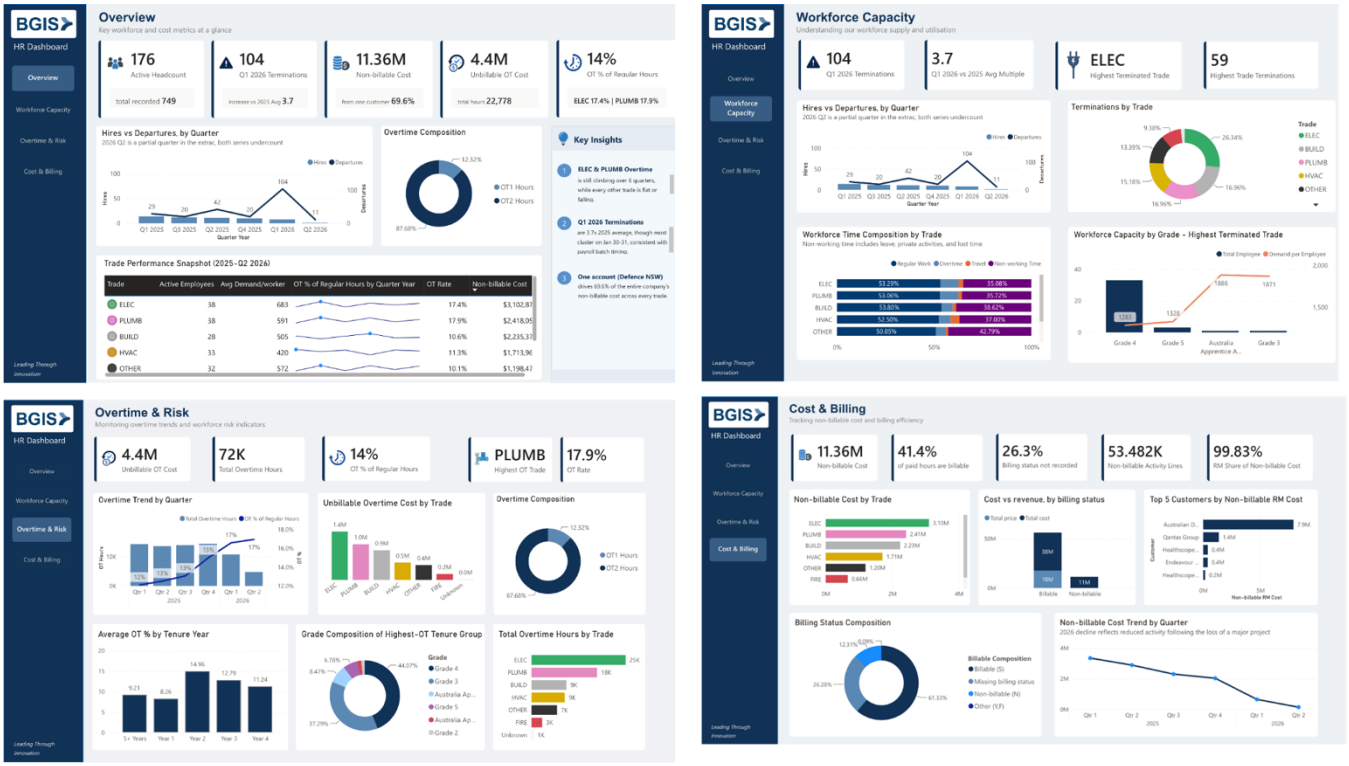


Figure 7. *HR Dashboard Recommendations*

The *Overview* page serves as the management entry point, answering the question *where should we pay attention first?* It brings the core workforce and financial signals together on one screen so a user can identify unusual movements or high-pressure areas before deciding where to drill down.

From there, the *Workforce Capacity* page addresses the resourcing question: *do we have the right workforce in the right areas?* It connects employee terminations with how time and workload are distributed, then breaks this down further by grade, supporting more targeted recruitment and workforce allocation decisions.

The *Overtime & Risk* page focuses on where workload pressure may be becoming costly or unsustainable. It shows how much overtime is being used, how heavily different groups rely on it, and how much of that overtime cost goes unrecovered, broken down by trade, tenure and grade. This lets HR distinguish whether overtime is a broad workforce issue or concentrated within particular groups.

Finally, the *Cost & Billing* page addresses the commercial impact directly: *what is driving non-billable cost, and where are the biggest recovery opportunities?* It identifies where non-billable cost is concentrated and surfaces the gaps in billing classification that make these costs difficult to track in the first place.

The dashboard is designed to update dynamically as new data is added, so the trades and workforce groups it flags as priorities can shift over time rather than being fixed to the findings in this report. It should, however, be read as a strong foundation rather than a complete diagnostic tool. Several questions central to workforce strategy cannot yet be answered reliably from the current data. In particular,

whether pay levels influence turnover, the point at which turnover begins to affect productivity, and how long new employees take to reach full productivity. Answering these would require data BGIS does not currently provide to the dashboard: documented exit reasons, external salary benchmarks, and a clearer, agreed measure of productivity. Expanding the dashboard to incorporate these inputs as they become available is the natural next iteration of this work (see Section 6, Future Opportunities).

4.4 KPIs & Expected Impact

Table 5. *Performance KPIs and Target Outcomes*

KPI	Baseline	Target direction
Non-billable share of workforce time	42.6%	↓ Decrease
Missing billing status	26.3%	↓ Decrease
Financial match rate	72.5%	↑ Increase
Unmatched employee records	350 / 749	↓ Decrease
Dashboard visibility & adoption	Not yet built	Real-time visibility; leadership adoption

5. Evaluation: Success, Limitations & Future Opportunities

Project success

The project succeeded in moving BGIS from an unverified internal narrative ("lowest turnover in a decade") to an evidence-based, quantified picture of where margin is being lost, and critically, in narrowing a company-wide-looking problem down to a specific, addressable root cause. Identifying that 69.6% of non-billable cost sits in one account and 99.83% sits in one work-type (RM vs PPM) converts a diffuse governance problem into a scoped, fundable fix, which is the standard this kind of engagement should be judged against.

Limitations

- The termination and non-billable figures are based on matched employee records only; the 350 of 749 people not yet matched to the Employee Pay Report mean true company-wide totals could differ once reconciled.
- Price Book Data cannot independently support margin analysis in which 99.8% of Unit Cost values are effectively zero so any dollar-margin claim depends on the Technician Type Costing crosswalk, which was hand-authored rather than key-joined and should be peer-reviewed before being used in client-facing financial commitments.
- The apparent decline in non-billable cost around the Defence contract's end date is a timing correlation observed in the data, not a confirmed causal finding, and is explicitly flagged for BGIS to confirm.
- A 5,230-row synchronisation gap between systems remains uninvestigated and could affect the precision of match-rate figures reported in Section 3.3.

Future opportunities

- Once Stage 2 data governance work resolves the ID and taxonomy issues, the same methodology could be extended to other client accounts beyond Defence-NNSW to check whether similar concentration patterns exist elsewhere in the book.
- The HR Interactive Dashboard's four live views create a foundation for predictive work; for example, flagging accounts trending toward a Defence-NNSW-style concentration before it reaches $11M.

6. Conclusion

This project delivered BGIS a clear, evidence-based account of how workforce vacancies, pay, underutilisation and capability gaps are affecting margin, anchored in a rigorously cleaned and validated five-dataset master file rather than assumption. The central contribution is diagnostic precision: rather than reporting that $11.36 million is non-billable, the analysis traces that figure to a single client account and a single work-type conversion problem, giving BGIS a fundable, sequenced path including contract review, data governance, then live visibility to act on immediately.

Group BW1 recommends BGIS treat the Stage 1 actions as time-sensitive given the scale of cost involved, while resourcing Stage 2 and 3 as the structural investment that prevents a comparable issue from going undetected again.

References

Cooke, J. (2026, July). Facilities management services in Australia (OD5528) [Industry report]. IBISWorld. https://www.ibisworld.com/

Jobs and Skills Australia. (2025). 2025 occupation shortage list: Key findings report. https://www.jobsandskills.gov.au/sites/default/files/2025-10/2025_osl_key_findings_report.pdf

Ritter, T., IFMA IT Community, Autodesk, & Clarke, S. (2024, January 22). Optimizing building management with a lifecycle approach. IFMA Knowledge Library. https://knowledgelibrary.ifma.org/optimizing-building-management-with-a-lifecycle-approach/

APPENDIX A. Dashboard Extracts (Supporting Documentation)

The following extracts from the team's Power BI HR dashboard support the findings reported in Section 4.

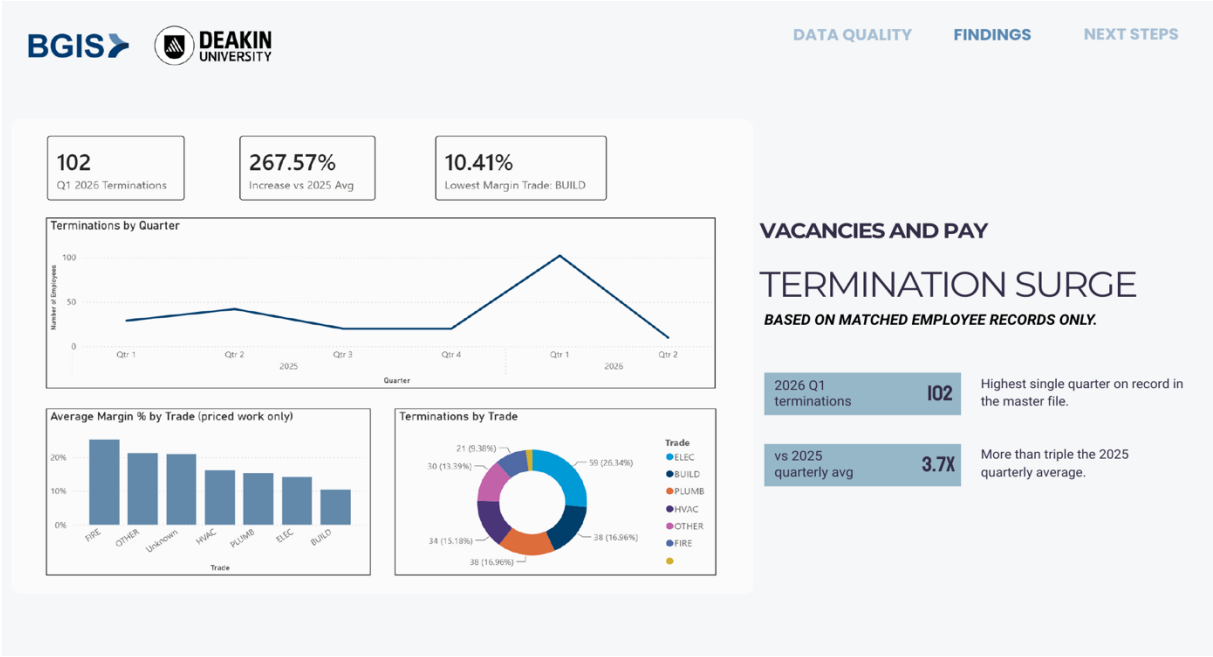


Figure A1. *Vacancies & Pay: termination surge and margin-by-trade view*

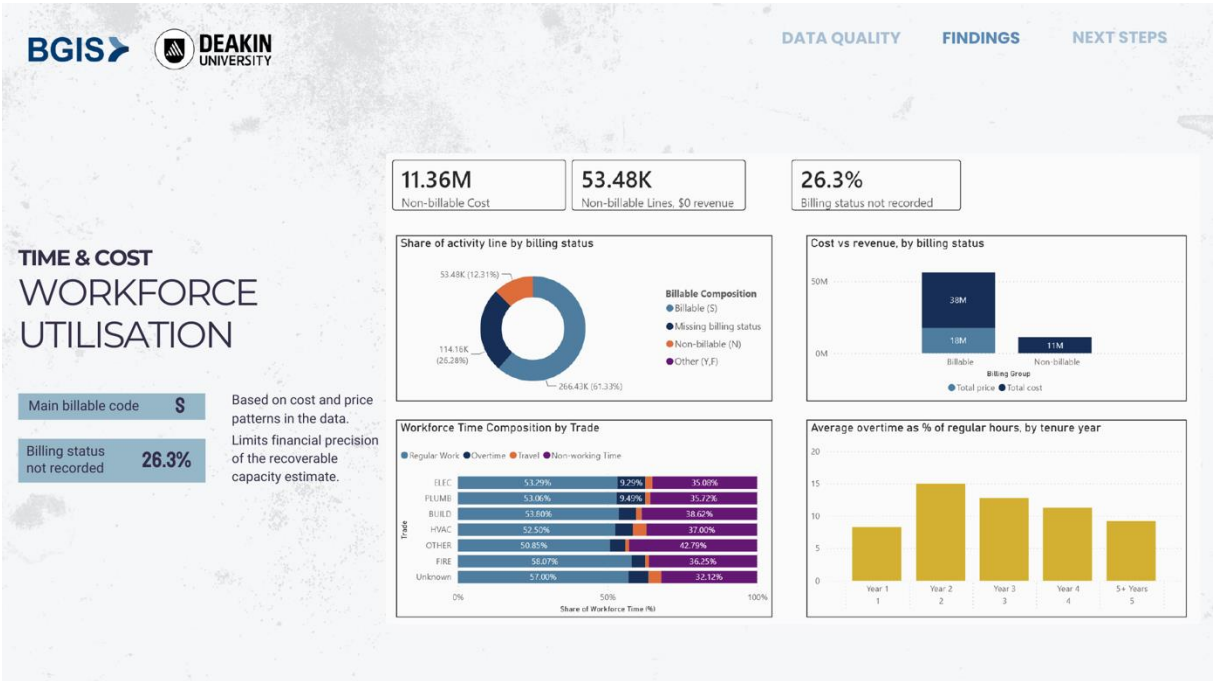


Figure A2. *Time & Cost: workforce utilisation view*

APPENDIX B. HR Interactive Dashboard (Supporting Documentation)

The HR Interactive Dashboard (Stage 3, Section 5.3) is built on BGIS's existing IFS Cloud and Power BI environment and is organised into four role-oriented pages, each mapping directly onto one of BGIS's original business problems: workforce overview, workforce capacity, overtime and risk, and cost and billing. The four pages are reproduced in full below as evidence of the analytical rigour and live visibility underpinning the recommendations in Section 5.

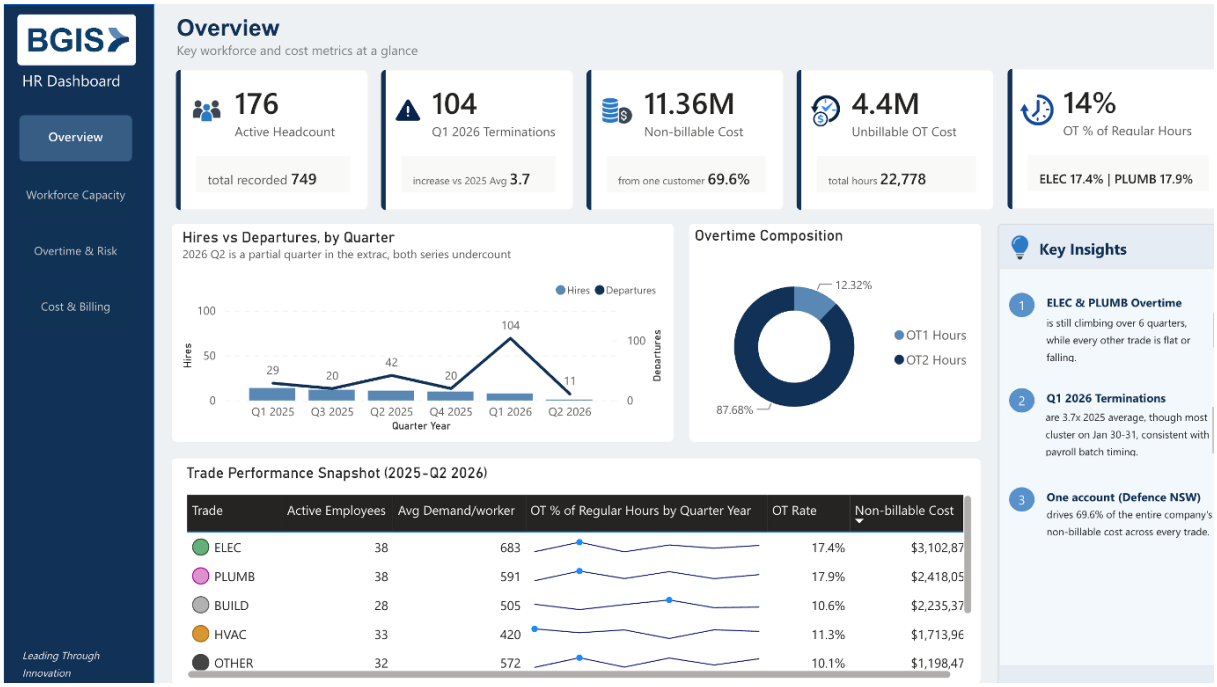


Figure B1. *Overview: headcount, terminations, non-billable cost and key insights at a glance*

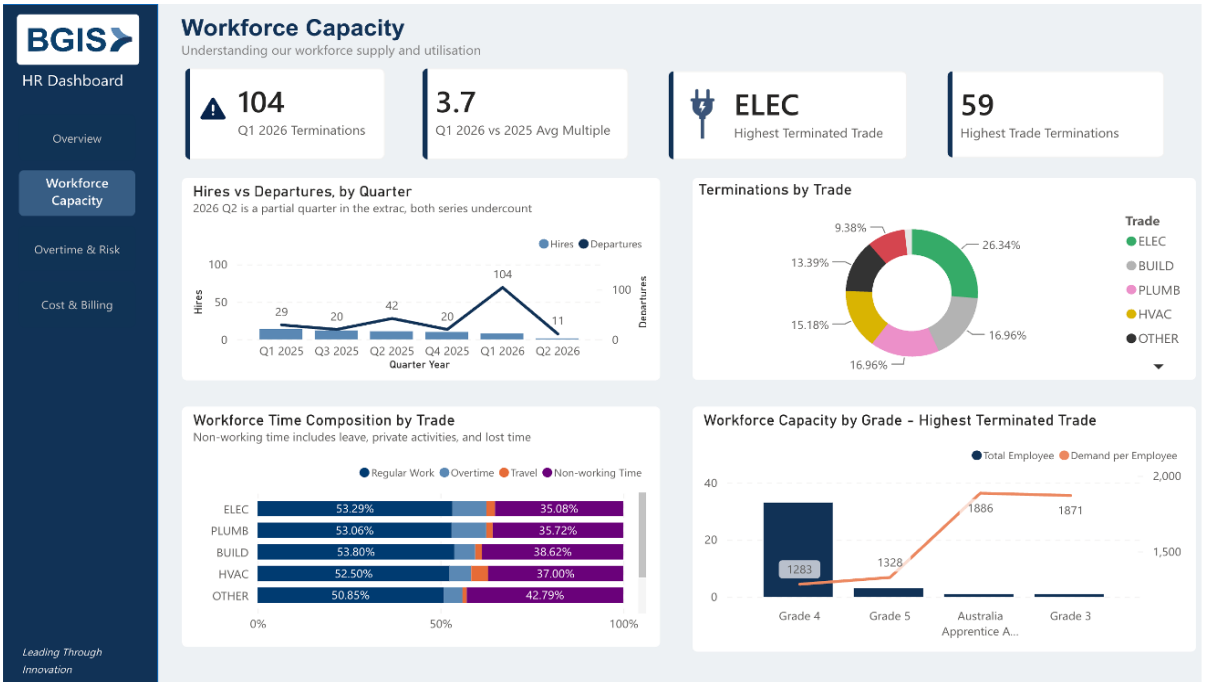


Figure B2*. Workforce Capacity: hires vs departures, terminations by trade, and workforce capacity by grade*



Figure B3. *Overtime & Risk: overtime trend, unbillable overtime cost by trade, and tenure-based OT patterns*



Figure B4. *Cost & Billing: non-billable cost by trade, cost vs revenue, and the Defence–NNSW account concentration*

APPENDIX C. Data Quality & Assumptions Summary

- "Unknown" Trade values are a retained data-quality category, not an inferred value.
- Cost and price fields from Financial Demands are reported as-is; "margin" language requires Finance validation before use.
- Price Book Data cost coverage is effectively zero for 99.8% of rows and cannot independently support margin analysis.
- price_book_candidate_match = True indicates a possible pricing combination exists, not that a specific price has been selected - Header/Line IDs and effective dates are still required.
- 114,157 rows previously counted as unmatched are Private Activities, not genuine matching failures, and should be tagged and excluded at entry going forward.
- A 5,230-row synchronisation gap between systems remains open and uninvestigated.

APPENDIX D. Master File Reference

Three audit files underpin all findings in this report and are available for BGIS review:

Table D1. *Master File Audit Outputs*

File	Purpose
MIS399_master_merged_audit.csv	Main activity-level checking file (Astea activity-row grain).
relationship_quality.csv	Match percentages across all dataset relationships.
unmatched_utilisation_to_financial.csv	Unique Order + Demand + Person keys in Astea with no Financial Demands match, retained for investigation.

APPENDIX E. DATA SOURCES AND MERGED DATA FILES

https://github.com/ashlynn-05/MIS399_DATA

APPENDIX F. MEETING MINUTES

Meeting Minutes 1:

MEETING MINUTES

Meeting/ Project Name:	**BW1 Meeting 1 – MIS399**		
Date: (MM/DD/YY)	27/07/2026	**Time:**	11:00 – 11:40 AM
Minutes Prepared By:	Thuong Luong	**Location:**	Online via Teams

1.Meeting Objective

- Make sure all team members understand what each dataset represents.
- Team agreement on the initial analytical structure.
- Task allocation and next steps.
- Team Planner completion.

2. Attendance

Attendees	S-number	Reason of Absence
Thuong Luong	223756642	N/A
Jade Dang	224323537	N/A
Quynh Hoang	224323537	N/A
Yasiru Dinara Abeyawardena	225551973	N/A
Shenal Nathan Witharana	225270297	N/A
Divya Hasandi Hewawickrama	225381863	N/A
Lynn Le	223856755	Uni schedule from 10AM-1PM

3. Content

- The team reviewed the datasets together and decided to split the work into pairs for further analysis.
- There are some columns and terms in the datasets that need further clarification so all questions will be noted down for consultant on the next seminar session (30/07).
- Team members will add more onto the team planner and allocate tasks to themselves.

4. Tasks Allocation

Tasks	Take Responsibility	Deadlines (Date & Time)	Note
Employee Pay Report initial analysis & cleaning	Yasiru, Divya	30/07	• Try to come up with potential insights/analysis themes. • Create a copy, try to play around and note down data cleaning steps for team approval.
Financial Demands initial analysis & cleaning	Jade, Quynh	30/07	
Price Book Data initial analysis & cleaning	Thuong, Lynn	30/07	
Astea Utilisation Data	Shenal, Yasiru	30/07	
Technician Type Costing review and analysis	All team members	30/07	
Team Planner add-ons	All team members	30/07	N/A

5. Uncomplete Tasks / Improvements

The meeting started at 11am but it took around 10 mins for everyone to show up. Hopefully members will be more punctual in our next meetings.

Meeting Minutes 2:

MEETING MINUTES

Meeting/ Project Name:	**BW1 Meeting 2 – MIS399**		
Date: (MM/DD/YY)	03/08/2026	**Time:**	1:00 – 1:40 AM
Minutes Prepared By:	Jade Dang	**Location:**	Online via Teams

1.Meeting Objective

- Update team member's progress
- Stay focused on data cleaning

2. Attendance

Attendees	S-number	Reason of Absence
Thuong Luong	223756642	N/A
Jade Dang	224323537	N/A
Quynh Hoang	224323537	N/A
Yasiru Dinara Abeyawardena	225551973	N/A
Shenal Nathan Witharana	225270297	N/A
Divya Hasandi Hewawickrama	225381863	Clashed class
Lynn Le	223856755	N/A

3. Content

- The team reviewed the datasets and continued data cleaning and preparation.
- Inconsistent data formats and values were identified across the datasets, requiring further investigation before analysis could begin.
- The team checked whether the available variables were relevant to the planned analysis and discussed how the data could be used to address the project objectives.
- Team members shared their progress and discussed outstanding issues that required further clarification.
- The team agreed to continue cleaning and reviewing the data before the next class so that unresolved issues could be raised with Fran.
- The team discussed the importance of establishing a consistent approach to data preparation before progressing to further analysis.

4. Tasks Allocation

Tasks	Take Responsibility	Deadlines (Date & Time)	Note
Continue data cleaning and preparation	All team members	06/08	Clean and review the datasets before the next class. Ensure available data supports the planned analysis. Bring unresolved data issues to the next class.
Review inconsistent data and identify unresolved issues	All team members	06/08	
Document key data-cleaning issues and questions	All team members	07/08	
Share progress and outstanding issues with the team	All team members	Before next meeting	

5. Uncomplete Tasks / Improvements

- Data cleaning was not fully completed due to inconsistencies across the datasets and required further review.
- Some data-related issues remained unresolved and were to be discussed with Fran during the next class.
- The meeting started at 1:00 PM, but it took approximately seven minutes for all available members to join. The team agreed to improve punctuality and start meetings more efficiently in future.
- The team should continue documenting data issues as they arise to avoid repeating the same investigation across subgroups.

Meeting Minutes 3:

MEETING MINUTES

Meeting/ Project Name:	**BW1 Meeting 3 - MIS399**		
Date: (MM/DD/YY)	24/08/2026	**Time:**	7:57 – 8:47 PM
Minutes Prepared By:	Divya Hewawickrama	**Location:**	Online via Teams

1.Meeting Objective

- Finalise and refine the Presentation 1 content and key insights.
- Review the data analysis, dashboards and presentation structure before presenting.

2. Attendance

Attendees	S-number	Reason of Absence
Thuong Luong	223756642	N/A
Jade Dang	224323537	N/A
Quynh Hoang	224323537	Work shift extended
Yasiru Dinara Abeyawardena	225551973	N/A
Shenal Nathan Witharana	225270297	N/A
Divya Hasandi Hewawickrama	225381863	N/A
Lynn Le	223856755	N/A

3. Content

- Discussed changing the KPIs under Workforce Utilization dashboard to provide more important and relevant insights.
- Discussed an issue related to billable hours.
- Listed the key points identified with the Termination dashboard.
- Discussed which insights should be shared at the end of Presentation 1 to create a smooth transition into Presentation 2.
- Gathered final thoughts on how the data was linked and compared how the percentages in the cleaned versions matched the original data, using Astea as the key source.
- Decided to divide the presentation slides among the presenting members after the final touches.

4. Tasks Allocation

Tasks	Take Responsibility	Deadlines (Date & Time)	Note
Data & Dashboard Review	All team members	26/08	Recheck KPIs, calculations and dashboard issues before presenting.
Presentation Refinement	All team members	26/08	Review and finalise the slides, visuals and key insights.
Practice the presentation	Yasiru, Divya and Shenal	26/08	Practice assigned slides and ensured a smooth flow between speakers.
Final Presentation Check	All team members	26/08	Check slide consistency, timing and overall presentation flow.

5. Uncomplete Tasks / Improvements

Finalise the Workforce Utilisation KPIs dashboard.

Complete the final presentation slides, including key insights, market research and the transition to Presentation 2.

A final check with the data links and calculations against the original datasets.

Meeting Minutes 4:

MEETING MINUTES

Meeting/ Project Name:	**BW1 Meeting 4 - MIS399**		
Date: (MM/DD/YY)	31/08/2026	**Time:**	4h-4h50 PM
Minutes Prepared By:	Quynh Hoang	**Location:**	Online via Teams

1.Meeting Objective

- Discuss on Presentation 1 feedback
- Plan for Presentation 2

2. Attendance

Attendees	S-number	Reason of Absence
Thuong Luong	223756642	N/A
Jade Dang	224323537	N/A
Quynh Hoang	224323537	N/A
Yasiru Dinara Abeyawardena	225551973	N/A
Shenal Nathan Witharana	225270297	N/A
Divya Hasandi Hewawickrama	225381863	N/A
Lynn Le	223856755	N/A

3. Content

- Reviewed Presentation 1 feedback and discussed the direction for Presentation 2.
- Discussed identifying additional data patterns to strengthen the insights for Presentation 2.
- Started building the interactive Power BI dashboard for Presentation 2, noting the dashboard should visibly support the team's recommendations, not just displaying raw data.
- Noted that the HR dashboard is the primary view the client engages with, so this should be prioritized in the building.
- Confirmed the Technician Type Costing and Price Book datasets have not yet been explored, used in the analysis so far
- Discussed structuring recommendations around three areas: (1) how the team handles missing/incomplete data, (2) how we deal with data gap/quality and (3) a plan or predictive-model direction for workforce utilization.
- Revisited the data merging approach, as the team noticed another group reported different match-rate numbers for similar joins; agreed to test alternative merging methods to see if the match rate can be improved.
- Discussed ideas to include in the written report in addition to the dashboard.

4. Tasks Allocation

Tasks	Take Responsibility	Deadlines (Date & Time)	Note
Explore Technician Type Costing & Price Book datasets	All team members	Before Wed 8PM	Identify margin/pricing insights not yet covered
Re-test data merging methods	All team members	Before Wed 8PM	Dashboard should visibly support the proposed recommendations
Draft recommendation for handling missing data	All team members	Before Wed 8PM	Compare against other group's match-rate numbers to validate accuracy
Draft workforce utilizations plan/model	All team members	Before Wed 8PM	For inclusion in report and dashboard
Draft the overall pathway for the HR interactive dashboard	All team members	Before Wed 8PM	Explore predictive model or action-plan direction

5. Uncomplete Tasks / Improvements

Data merging approach needs re-validation against alternative methods (if possible)

Meeting Minutes 5:

MEETING MINUTES

Meeting/ Project Name:	**BW1 Meeting 5 - MIS399**		
Date: (MM/DD/YY)	2/9/2026	**Time:**	8:31-9:02 PM
	3/9/2026	**Time:**	10:05-12:29 AM
	6/9/2026	**Time:**	4:04-4:31 PM
Minutes Prepared By:	Thuong Luong	**Location:**	Online via Teams

1.Meeting Objective

- Review the draft of Presentation 2 before the mentor consultation.
- Check the flow between initial findings, further analysis and recommendations.
- Identify areas requiring mentor feedback and further improvement.

2. Attendance

Attendees	S-number	Reason of Absence
Thuong Luong	223756642	N/A
Jade Dang	224323537	N/A
Quynh Hoang	224323537	N/A
Yasiru Dinara Abeyawardena	225551973	N/A
Shenal Nathan Witharana	225270297	N/A
Divya Hasandi Hewawickrama	225381863	N/A
Lynn Le	223856755	N/A

3. Content

- Reviewed the overall Presentation 2 draft structure and storyline.
- Checked whether the initial findings from Presentation 1 were clearly connected to further analysis.
- Reviewed the additional findings developed around ELEC and PLUMB, non-billable cost, workforce activity and termination patterns.
- Discussed whether further findings provided sufficient evidence to support the proposed recommendations.
- Reviewed the proposed three-stage recommendation roadmap.
- Checked whether the Data Quality & Governance recommendation addressed the main data limitations identified during the analysis.
- Reviewed the progress of the HR Interactive Dashboard and whether its workforce, overtime, cost and billing views supported the presentation findings.
- Checked figures, terminology, visual consistency and slide flow before the mentor consultation.
- Identified areas of uncertainty and questions to raise with the mentors before finalising Presentation 2.

4. Tasks Allocation

Task	Responsibility	Deadline	Note
Refine Presentation 2 draft	All team members	Before mentor consultation	Improve storyline and slide flow
Validate findings and figures	All team members	Before mentor consultation	Check against final analysis
Review recommendations and dashboard	All team members	Before mentor consultation	Ensure recommendations connect to findings
Prepare questions for mentors	All team members	Before mentor consultation	Identify areas requiring feedback

5. Uncomplete Tasks / Improvements

Presentation 2 recommendations, dashboard content and slide structure required further refinement following mentor feedback.

Meeting Minutes 6:

MEETING MINUTES

Meeting/ Project Name:	**BW1 Meeting 6 - MIS399**		
Date: (MM/DD/YY)	14/09/2026	**Time:**	8:30 - 9:19 PM
Minutes Prepared By:	Yasiru Abeyawardena	**Location:**	Online via Teams

1.Meeting Objective

- Review the completed Presentation 2 slide set with the presenting members.
- Allocate presentation sections based on each presenter's confidence and preferred content.
- Confirm the overall presentation flow and speaker handovers.
- Identify any areas requiring further preparation before the full-team rehearsal.

2. Attendance

Attendees	S-number	Reason of Absence
Thuong Luong	223756642	Present
Jade Dang	224323537	Present
Quynh Hoang	224323537	Present
Yasiru Dinara Abeyawardena	225551973	Not required - non-presenting team member
Shenal Nathan Witharana	225270297	Not required - non-presenting team member
Divya Hasandi Hewawickrama	225381863	Not required - non-presenting team member
Lynn Le	223856755	Present

3. Content

- Reviewed the full Presentation 2 slide set and discussed the purpose of each section.
- Confirmed that **Thuong, Lynn, Quynh and Jade** would be the four presenters for Presentation 2.
- Discussed which findings and recommendations each presenter felt most confident presenting.
- Divided the presentation slides among the four presenters based on familiarity with the content and preferred speaking sections.
- Reviewed the expected sequence from initial findings to further insights and final recommendations.
- Discussed speaker transitions to ensure the presentation would feel continuous rather than like four separate sections.
- Identified sections that required individual practice before the next rehearsal.
- Agreed that the full team would review the presentation and provide feedback during the next meeting.

4. Tasks Allocation

Tasks	Take Responsibility	Deadlines (Date & Time)	Note
Review assigned slides	Thuong, Lynn, Quynh, Jade	16/09	Become familiar with assigned findings and recommendations.
Prepare individual speaking points	Presenting members	16/09	Avoid reading directly from slides and maintain concise explanations.
Review speaker transitions	Presenting members	16/09	Ensure smooth handovers between presentation sections.
Prepare full-team rehearsal	All team members	16/09	Presenters rehearse; remaining members provide feedback.

5. Uncomplete Tasks / Improvements

- Presenters to practise their allocated sections individually.
- Confirm total presentation timing during the next rehearsal.
- Review difficult or information-heavy slides with the full team.
- Finalise any wording or visual changes after receiving team feedback.

Meeting Minutes 7:

MEETING MINUTES

Meeting/ Project Name:	**BW1 Meeting 7 - MIS399**		
Date: (MM/DD/YY)	16/09/2026	**Time:**	8:47 – 11:43 PM
Minutes Prepared By:	Yasiru Abeyawardena	**Location:**	Online via Teams

1.Meeting Objective

- Conduct a complete rehearsal of Presentation 2 before submission.
- Evaluate timing, clarity, speaker transitions and overall presentation flow.
- Allow non-presenting members to provide feedback on the presentation.
- Make final amendments to the slide deck before submission.

2. Attendance

Attendees	S-number	Role
Thuong Luong	223756642	Presenter
Jade Dang	224323537	Presenter
Quynh Hoang	224323537	Presenter
Yasiru Dinara Abeyawardena	225551973	Feedback / review
Shenal Nathan Witharana	225270297	Feedback / review
Divya Hasandi Hewawickrama	225381863	Feedback / review
Lynn Le	223856755	Presenter

3. Content

- The four presenting members delivered the full presentation **three times**.

- Reviewed whether the presentation clearly progressed from:
- initial findings,
- further findings and insights,
- recommendations.
- Yasiru, Divya and Shenal listened to each rehearsal and provided feedback on:
- clarity of explanations,
- slide readability,
- speaker pacing,
- transitions between presenters,
- areas where the audience may have difficulty understanding the message.
- Reviewed the slide set after the rehearsals and identified areas that contained too much information or required clearer wording.
- Made adjustments to simplify difficult slides and improve the overall client-facing narrative.
- Checked that the recommendations clearly linked back to BGIS's original workforce problems.
- Reviewed whether the presentation adequately addressed feedback received from Presentation 1.
- Checked visual consistency across headings, fonts, colours and slide structure.
- Finalised the Presentation 2 deck following the amendments.
- Submitted the completed presentation slide deck through the assessment portal.

4. Tasks Allocation

Task	Responsibility	Deadline	Note
Complete full presentation rehearsal	Four presenters	16/09	Presentation rehearsed three times.
Provide presentation feedback	Yasiru, Divya, Shenal	16/09	Focus on clarity, flow, timing and client understanding.
Amend difficult slides	All team members	16/09	Simplify content and improve visual readability.
Final presentation consistency check	All team members	16/09	Check fonts, headings, colours, slide numbers and wording.
Submit Presentation 2 deck	Team	16/09	Final slide set submitted after amendments.
Complete final timing rehearsal	All team members	17/09	Confirm presentation remains within 10-minute limit.

5. Uncomplete Tasks / Improvements

- Conduct one final rehearsal on the morning of the client presentation.
- Confirm the presentation remains within the required **10-minute limit**.
- Presenters to review their final speaking points and transitions.
- Avoid making unnecessary changes to the submitted slide content unless essential.

Meeting Minutes 8:

MEETING MINUTES

Meeting/ Project Name:	BW1 Meeting 8 - Final Presentation 2 Rehearsal		
Date: (MM/DD/YY)	17/09/2026	Time:	8:59 – 9:58AM
Minutes Prepared By:	Yasiru Abeyawardena	Location:	Online via Teams

1.Meeting Objective

- Complete the final rehearsal immediately before Presentation 2.
- Confirm the presentation fits within the required 10-minute time limit.
- Check speaker pacing, transitions and confidence.
- Resolve any final delivery issues before presenting to the client.

2. Attendance

Attendees	S-number	Role
Thuong Luong	223756642	Presenter
Jade Dang	224323537	Presenter
Quynh Hoang	224323537	Presenter
Yasiru Dinara Abeyawardena	225551973	Feedback / timing review
Shenal Nathan Witharana	225270297	Feedback / timing review
Divya Hasandi Hewawickrama	225381863	Feedback / timing review
Lynn Le	223856755	Presenter

3. Content

- The four presenting members delivered the complete Presentation 2 deck in presentation order.
- The full rehearsal was timed to confirm compliance with the 10 minute presentation limit.
- The final rehearsal finished just under 10 minutes, confirming that the presentation could be delivered within the required timeframe.
- Yasiru, Divya and Shenal listened to the rehearsal and provided final comments on pacing, clarity and transitions.
- Confirmed that presenters understood their allocated sections and handover points.
- Reviewed whether any sections were being explained too quickly or taking unnecessary time.
- Confirmed that the most important findings and recommendations were receiving sufficient presentation time.
- Avoided major last-minute slide changes to maintain presenter confidence and consistency.
- Confirmed the team was ready for the 10:00 AM client presentation.

4. Tasks Allocation

Task	Responsibility	Deadline	Note
Deliver final timed rehearsal	Thuong, Lynn, Quynh, Jade	Before 10:00 AM	Completed just under the 10-minute limit.
Monitor timing	Yasiru / team	Before presentation	Ensure overall presentation remains within time.
Final delivery feedback	Yasiru, Divya, Shenal	Before presentation	Focus on pace, clarity and handovers.
Deliver Presentation 2	Presenting members	10:00 AM	Present final findings and recommendations to client.
Support Q&A / discussion	All team members	Following presentation	Assist where clarification is required.

5. Uncomplete Tasks / Improvements

- Deliver Presentation 2 to the client.
- Record any client feedback or requested clarification following the presentation.
- Carry relevant Presentation 2 feedback forward into the final project report.

Meeting Minutes 9:

MEETING MINUTES

Meeting/ Project Name:	**BW1 Meeting 9 - MIS399**		
Date: (MM/DD/YY)	9/2026	**Time:**	
Minutes Prepared By:	Lynn Le	**Location:**	Online via Teams

1.Meeting Objective

- Plan the structure and content of the final report.
- Allocate report sections and final project tasks among team members.

2. Attendance

Attendees	S-number	Reason of Absence
Thuong Luong	223756642	N/A
Jade Dang	224323537	N/A
Quynh Hoang	224323537	N/A
Yasiru Dinara Abeyawardena	225551973	N/A
Shenal Nathan Witharana	225270297	N/A
Divya Hasandi Hewawickrama	225381863	N/A
Lynn Le	223856755	N/A

3. Content

- Reviewed the feedback received following Presentation 2 and discussed improvements to be reflected in the final report.
- Discussed how the findings from Presentation 1 and Presentation 2 should be combined into one clear report storyline.
- Agreed on the overall structure of the final report
- Discussed expanding the presentation findings with more detailed explanations, evidence and supporting figures for the written report.
- Agreed to document the initial data quality gaps separately before presenting the Data Quality & Governance recommendation.
- Discussed incorporating the HR Interactive Dashboard into the report as the proposed ongoing workforce monitoring solution.
- Planned to bring all report sections together into one complete draft for final checking and improvements.

4. Tasks Allocation

Task	Responsibility	Deadline	Notes
Write data analysis approach	All team members	Before report compilation	Explain the analysis and data preparation process
Write BGIS overview & market research			Provide relevant company and industry context
Write initial data quality gaps			Document key issues identified during analysis
Write Recommendation 1: Data Quality & Governance			Link identified data gaps to proposed improvements
Write Recommendation 2: HR Interactive Dashboard			Explain the dashboard solution and implementation
Compile supporting documents and appendices		Before final review	Organise supporting evidence and project materials
Combine and format final report			Ensure consistent structure, formatting and referencing

5. Uncomplete Tasks / Improvements

Referencing, formatting, figures, appendices and overall consistency will be checked once the full report draft is compiled.